

Index Laws - Review

$$a^m \times a^n = a^{m+n}$$

$$a^m \div a^n = a^{m-n}$$

$$(a^m)^n = a^{m \times n}$$

$$a^0 = 1, \quad a \neq 0$$

Examples - Simplify as a single power. Show working.

a) $4^3 \times 4^8$

e) $\frac{5^2 \times 5^7 \times 5^{10}}{5^9 \times 5^2 \times 5^3}$

f) $\frac{(6^4)^5 \times (6^7)^0}{(6^4)^4}$

b) $\frac{7^9}{7^6}$

c) 20^0

d) $(12^3)^5$